

Lakshmi Swarna Durga Nallam

swarnadurga.nallam@gmail.com | (713) 359-6070 | [linkedin.com/in/swarna-nallam](https://www.linkedin.com/in/swarna-nallam) | [Portfolio](#)

EDUCATION

Rice University , Houston, United States - <i>Master's, Data Science and Machine Learning</i> - GPA: 3.73/4	Aug 2024 - Dec 2025
SRM University , Amaravati, India - <i>Bachelor, Computer Science and Engineering</i> - GPA: 8.55/10	Aug 2019 - May 2023

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, C/C++, Java
Machine Learning & AI: Google ADK, LLMs, RAG, Prompt Engineering, Recommendation Systems, NLP, Feature Engineering
Libraries / Models: Scikit-Learn, PyTorch, TensorFlow, BERT, XGBoost, LightGBM, CatBoost, Autoencoders
Data Engineering & Platforms: Hive, AWS, GCP, Vertex AI, Databricks, Apache Spark, PySpark, Hadoop, MongoDB, Git
Analytics and Visualization: Tableau, Microsoft Power BI, Matplotlib
Web Technologies: HTML, CSS, Bootstrap, JavaScript, TypeScript

WORK EXPERIENCE

AI Engineer , Rice University, Houston, Texas	Jun 2025 - Aug 2025
- Designed and deployed an end to end AI automation platform dmv-agent.ai, for vehicle registration workflows, integrating OCR and LLM pipelines to reduce manual processing time from 2+ hours to under 10 minutes. - Architected a real time document processing pipeline using TrOCR and Phi-based LLMs, with structured data extraction, validation layers, and MongoDB storage achieving 95% accuracy. - Developed RESTful APIs (FastAPI) and integrated AWS-based deployment, enabling scalable document ingestion, asynchronous processing, and seamless interaction between frontend and AI services. - Designed a human-in-the-loop decision workflow with fallback logic, enabling adaptive handling of low-confidence predictions and improving system reliability.	
Teaching Assistant , Rice University, Houston, Texas	Jan 2025 - May 2025
- Supported graduate-level NLP instruction on neural models, sentiment analysis, and information extraction for a 30-student cohort. - Led CUDA-based optimization labs, mentoring students on GPU acceleration and fine-tuning transformer architectures. - Provided technical guidance on neural architectures and model performance trade offs, reinforcing practical system design.	
Full Stack Developer , Barclays, Chennai, India	Jul 2023 - Jul 2024
- Developed scalable AEM 6.5 components and workflows using Java, Sling Models, OSGi, and JavaScript for 10+ US co-branded banking pages. - Integrated backend services(FastAPI) and CI/CD pipelines(Jenkins, Maven, Git) to automate build, testing and deployments across environments. - Collaborated with cross-functional teams to optimize DAM workflows, performed 50+ code reviews, and improve application reliability, maintainability, and release efficiency in Agile delivery cycles.	
Data Science Intern , APSSDC, Vijayawada, India	Mar 2022 - Aug 2022
- Built CLTV, churn prediction, and segmentation models using pandas, scikit-learn, and SQL to drive marketing and loyalty strategies. - Designed ARIMA-based demand forecasting and frequency-reward systems to improve store-level planning and customer retention.	
Student Researcher , SRM University, Amaravati, India Link	Jun 2021 - Jul 2021
- Analyzed COVID-19 epidemiological data using a multivariate linear regression model comparing India's first and second waves; observed daily cases increased significantly from 30,000 (first wave) to 120,000 (second wave) due to the Delta variant. - Co-authored Springer publication with visual analytics on population density, policy impact, and fatality rate reduction (4% → 1%).	
OPEN SOURCE PROJECTS	
AutoApply AI SerpAPI, Google Gen AI SDK, Playwright, Python, Gemini Link	Feb 2026 - Mar 2026
- Designed a multi-agent AI workflow for automated job discovery, ranking, and application orchestration, combining SerpAPI, Gemini LLMs, and browser automation to process and prioritize technical roles in real time. - Built an LLM judge layer and human-in-the-loop application pipeline using Playwright, enabling adaptive autofill, persistent browser session management, secure login reuse, and reliable end-to-end execution. - Developed modular automation agents with recovery, fallback, and retry mechanisms, including standalone discovery pipelines and CSV-based ranking workflows, improving application throughput by 2–3×.	
E-commerce AI Data Analyst Agent Google ADK, Gemini, BigQuery, Cloud Run Link	Jan 2026 - Feb 2026
Developed a serverless LLM-powered analytics agent on GCP with schema discovery, function-calling orchestration, and real-time SQL generation pipelines to automate analytical reporting workflows across complex relational datasets.	
Travel ChatBot with RAG Framework LLM, Sentence Transformers, Python Link	Sep 2025 - Dec 2025
Designed a retrieval-augmented generation (RAG) pipeline over 30K+ UNESCO documents using FAISS and transformer embeddings, achieving 92% query accuracy, sub-1.2s latency, 78% storage reduction, and 41% improved retrieval performance over TF-IDF baselines.	